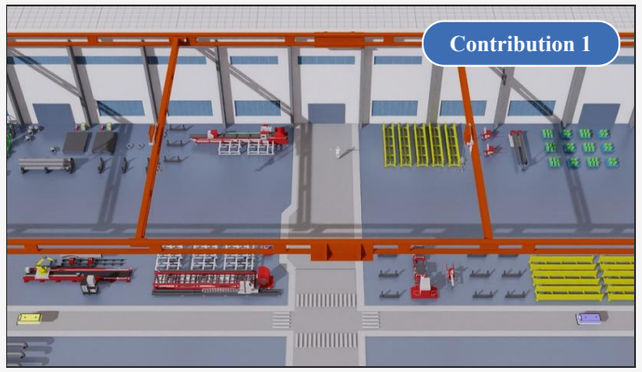
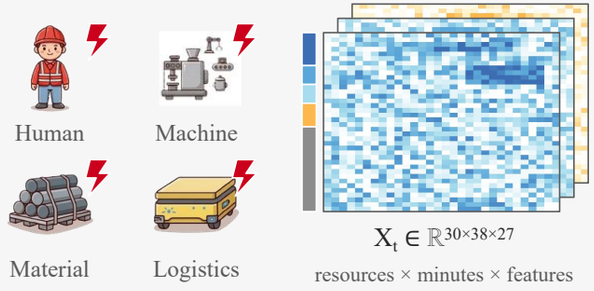
Digital twin & multi-entity data



Isaac Sim line · single-axis disturbances



$X_t \in \mathbb{R}^{30\times38\times27}$

resources × minutes × features

Contribution 1 · Multi-entity heterogeneous graph

✕ Prior work

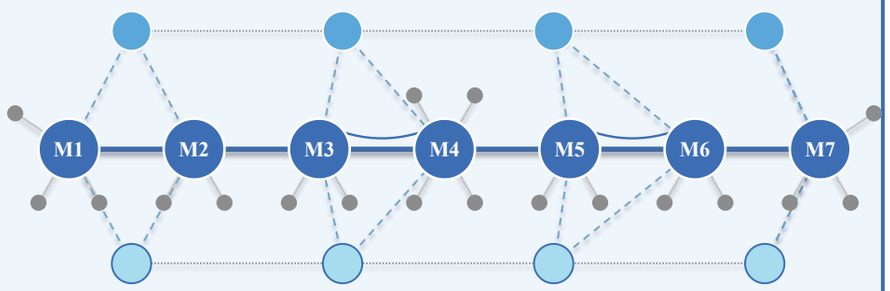
machine-only nodes

✓ PHGST-DP

5 entity types, typed edges

M^G: < 4-hop neighbours

M^S: top-5 trajectory correlation



machine · gantry · robot · human · buffer

process · handling · weak · buffer link

Contribution 2 · Process-informed event labels

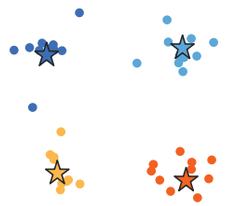
✕ Prior work

threshold spikes ≠ bottleneck

✓ PHGST-DP

sustained events, 8-min rule

Short-pattern clustering



16 pattern keys K^P

regimes, not labels

Type-specific rule b^{raw}

persistence ↓ m = 8, g = 1

Event target (w, s, d)

history · s · d

ongoing: hot at history end

upcoming: s > 0, d ≥ 8 min

+ cause: transport · shortage · starvation · queue buildup

pattern keys K^P

training targets

PHGST-DP

X_t

G, M^G, M^S

Grouped embedding

Contribution 1

Queue	4
Stall	6
Logistics	4
Shortage	1
Context	7
Type	5

mean

$+ E_{time} + E_{lap} + E_{node}$

$\rightarrow H^{(0)}$

Encoder block × 5 (d = 80)

Contribution 3 · Delay-propagation module

✕ Prior work

same-time links

✓ PHGST-DP

lagged precursors

Pattern attention

queue patch $P_{t,n}$

× 16 keys $K^P \rightarrow C^P$

C^P added to geo. keys

Geographic attn ×2 mask M^G, keys + C^P

Temporal attn ×4 per resource, 30 windows

Semantic attn ×2 mask M^S, similar resources

Concat + project

Add & Norm + FFN

Multi-task heads

Occurrence p^{cont} / p^{onset}
station event

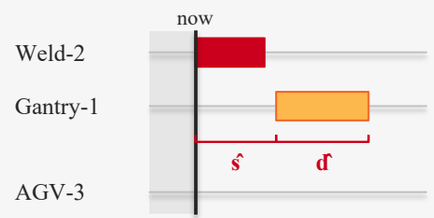
Start π^s · duration $\hat{d}$
when and how long

Prefix $\hat{p}$ · occupancy $\hat{O}$
dense future grid

Cause $\hat{c}$ · remaining $\hat{\ell}$
graph-level pooling

validation-controlled decoding

Bottleneck report



future minutes (cap 5/10/15)

Process cause $\hat{c}$

Transport · Shortage · Starvation · Queue

Remaining order time $\hat{\ell}$

12 min left